

HARSH KUMAR

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EDUCATION

M.S. in Artificial Intelligence, Columbia University Incoming, Fall 2026
B.Tech in Computer Science (Hons.), AI & ML, Manipal University Jaipur 2022-2026
Relevant Coursework: DSA, OOP, RDBMS, Artificial Intelligence, Soft Computing, Data Science, ML, NLP, Computer Vision, RL
CGPA: 8.95/10

SKILLS

Languages	Python, SQL (MySQL), TypeScript
ML & AI	LLMs, RAG, NLP, Deep Learning, Computer Vision, Multi-Agent Systems, Prompt Engineering, Fine-Tuning
Frameworks	PyTorch, TensorFlow, HuggingFace Transformers, LangChain, LangGraph, FastAPI, Scikit-Learn, OpenCV
Tools & Infra	Git, AWS Bedrock, ChromaDB, Qdrant, Ollama, Groq, vLLM, ServiceNow, MCP, CI/CD

EXPERIENCE

NTT Data — Intern (ServiceNow & Agentic AI) Dec 2025 - May 2026

- Built an **ITIL v4-compliant firewall change-management** solution on **ServiceNow** with Pre-Approved & Planned Change catalog items.
- Automated **multi-stage approval workflows** via **Flow Designer, Business Rules & GlideScript**, adding **RBAC, RITM** generation, and full audit trails.
- Prototyped an **Agentic AI ITSM platform (LangGraph, RAG, ChromaDB, FastAPI)** for incident classification and **evidence-grounded** resolution (see Projects).

Quest Global — AI Intern May 2025 - July 2025

- Built a **context-enriched scalable RAG pipeline** with multiple LLMs (**LLaMA, Mistral**) for test-case generation at **80–90%** accuracy, cutting manual effort by **70%+**.
- Engineered an autonomous test-execution **AI agent** using **AWS Strands SDK, Bedrock, Claude, MCP & Playwright** for fully autonomous validation.
- Delivered **AI-driven QA automation**, cutting manual workload by **60%** and accelerating testing cycles by **30%**.

Action Tour Guide — Python Developer Intern June 2024 - July 2024

- Built **Python automation** to extract, process & organize customer-email data, reducing manual effort by **40%**.
- Automated **post-tour follow-up email workflows**, boosting feedback response rates by **25%**.

PUBLICATIONS

Chronocept: Instilling a Sense of Time in Machines [🔗](#) Accepted at *EACL SRW 2026*

- Co-authored the **first benchmark** modeling **temporal validity** as a continuous probability distribution via **skew-normal curves** over a log-time axis.
- Built a **MATRES-grounded** annotation framework across **8 temporal axes**, curating **1,700+** samples at **84–89% IAA**.
- Benchmarked geometry-aware models (**RoBERTa, DeBERTa-v3, MT-DNN, SBERT**) under a **Gaussian NLL** objective predicting location, scale & skewness.

PermaFrost-Attack: Stealth Pretraining Seeding for Planting Logic Landmines [🔗](#) Under Review, *EMNLP 2026*

- First-authored a novel **web-scale poisoning** threat model, **Stealth Pretraining Seeding (SPS)**, planting dormant **“logic landmines”** fired by precise alphanumeric triggers.
- Designed **2 geometric diagnostics (Thermodynamic Length, and Infection Traceback Graph)** tracing adversarial influence invisible in model outputs.
- Validated persistent, triggerable deviations across **6 instruction-tuned LLMs (1B–14B)** via **SFT** on **HH-RLHF**, tested on in/out-of-distribution (**LITMUS**) prompts.

PROJECTS

AgentDesk — Agentic ITSM Triage Platform — (*LangGraph, RAG, ChromaDB, FastAPI, React*)

- Engineered a **4-agent LangGraph** pipeline (Intake → Retrieval → Resolution → Governance) with **ChromaDB RAG**, automating triage over **25,000+** incidents.
- Built a **dual-scoring guardrail** (cosine similarity + **LLM self-critique**) auto-resolving tickets at a **≥0.6** grounding score while escalating low-confidence cases with **100%** audit logging.
- Shipped a portable **FastAPI** backend on the **ServiceNow incident API** plus a **React/TypeScript** trace viewer, enabling **zero-code** mock-to-PDI migration.

Aegis — LLM Red-Teaming CLI — (*Python, asyncio, FastAPI, Click*)

- Built an async **Python CLI** red-teaming any LLM (**OpenAI, Anthropic, Ollama, HuggingFace, vLLM/Groq**) with **70+** attacks across **4** categories.
- Engineered a **YAML-driven** framework with **5× concurrency** and hybrid **regex + LLM-judge** scoring, producing graded **A–F** safety report cards.
- Shipped **multi-turn jailbreaks, attack chaining & differential testing** with **JSON/MD/HTML** reports and a **FastAPI** dashboard for **CI/CD** safety gating.

MedVision-LLM — AI Medical Diagnosis System — *(PyTorch, Ensemble DL, GPT-4, Streamlit)*

- Built an **ensemble model (EfficientNet-B3 + ResNet50 + DenseNet121)** for 4-class chest X-ray classification at **96.22%** accuracy across **10,000+** images.
- Added **Grad-CAM** explainability + **GPT-4** report generation, cutting interpretation time to **<2s** per case.
- Optimized with **focal loss, FP16 & cosine annealing (40% faster training)**; deployed a **Streamlit** app at **<2s** latency.

ACHIEVEMENTS AND CERTIFICATIONS

- 3 x Dean's List for Academic Excellence
- Machine Learning Specialization by DeepLearning.AI, Stanford by Andrew Ng.
- Finalist for National SAP Hackathon.
- Data Structures Certification from UC San Diego on Coursera
- CCNAv7 Cisco Certification on Switching, Routing & Wireless Essentials, and Enterprise Networking, Security and Automation
- Swayam NPTEL Course Certification for Machine Learning